SUPPORTING ADULT GUIDANCE

Inequalities as regions

Equations, Inequalities and Simultaneous Relationships · Lesson 11. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This closing lesson represents a linear inequality in two variables as a **region** (half-plane): a boundary line (dashed/solid by strictness) with the satisfying side shaded, chosen by a single test point. It extends to the **overlap** of two inequalities, linking explicitly to compound loci from Constructions, and consolidates the unit's widening idea of a 'solution' (value → pair → line → intersection → range → region).

Driving Question: *Where do all the allowed points live on a graph?*

The mathematics, in plain terms

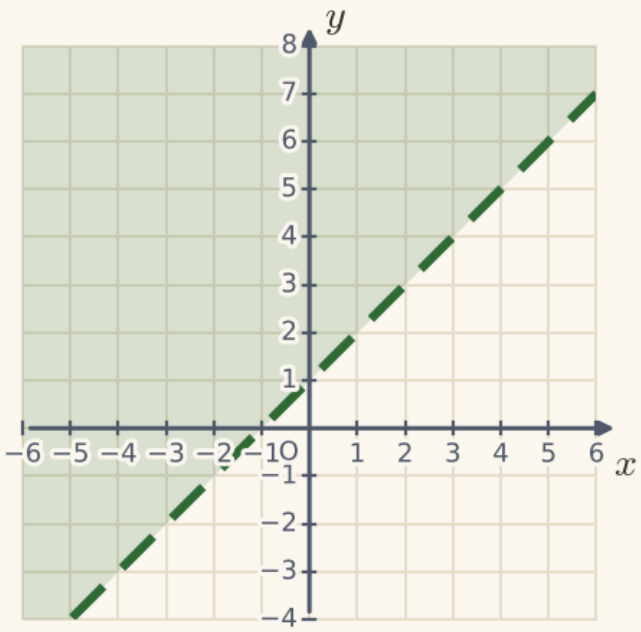
A linear inequality ax + by ≤ c (or $y \leq mx + c$, $x \geq a$, etc.) is satisfied by a half-plane bounded by the corresponding line. The line is included (solid) for non-strict inequalities and excluded (dashed) for strict ones. Substituting any point not on the line determines which half-plane satisfies it. The solution set of a system of inequalities is the intersection of the half-planes — the foundation of linear programming.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Shade a region

Shade the region $y > x + 1$.

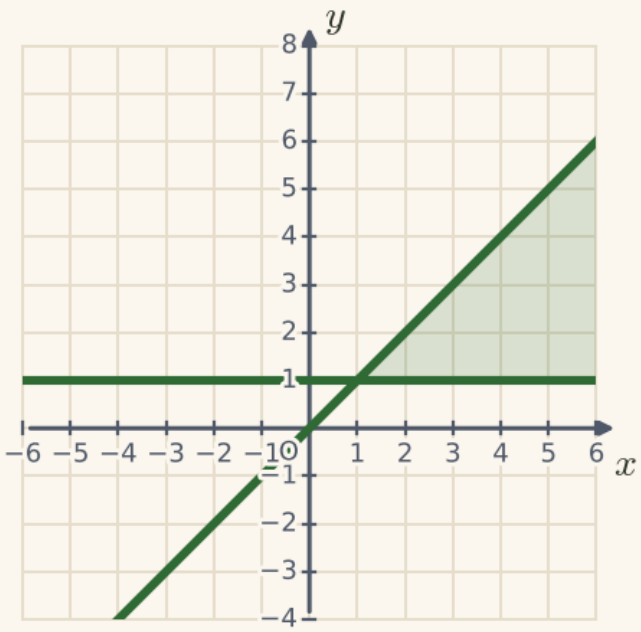


1. Draw the boundary $y = x + 1$, dashed (> is strict).	dashed line
2. Test the origin: is $0 > 0 + 1$?	$0 > 1$ is false
3. So shade the OTHER side — above the line.	region shaded above

$y > x + 1$ is the half-plane above the dashed line $y = x + 1$ (the origin is below it, and $0 > 1$ is false).

Example 2 — Where two regions overlap

Shade the region where $y \geq 1$ AND $y \leq x$.



1. $y \geq 1$: solid horizontal boundary, shade above.	above $y = 1$
2. $y \leq x$: solid boundary $y = x$, shade below.	below $y = x$
3. The answer is where both hold — the overlap.	the wedge between them

The overlap is the wedge where $1 \leq y \leq x$ — exactly the compound-region idea from Constructions, built from straight boundaries.

Boundary, then test, then shade

Keep the routine: draw and style the boundary (dashed/solid), pick a test point off the line, substitute to decide the side, shade. Reserve the origin as the default test point (fast, unless the line passes through it). For a system, shade each region lightly and identify the overlap. Emphasise checking with a point inside and one outside. Use the closing task to review how 'solution' has grown across the unit.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	See a region: boundary style and shaded side, with the origin test.	~9 min

Movement	Test the origin and choose which side to shade.	~10 min
Reflection	Build a region step by step: style → test → side.	~12 min
Creativity	Shade the overlap of two inequalities (the Constructions link).	~9 min
Rest	A short pause to close the unit and its widening idea of a solution.	~2 min
Nutrition	Intellectual: seeing all allowed possibilities at once — constraints and overlaps.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Solid boundary for a strict inequality (or vice versa).
Tie the line to the sign: '≤ or ≥ includes the boundary → solid; < or > excludes it → dashed.'

Shading the wrong side.
Return to the test point: 'Does the origin make the inequality true? Then shade its side; if not, the other side.'

Testing a point that lies on the boundary.
Choose a point clearly off the line. 'Is (0,0) on the line? If so, pick another test point, like (1,0).'

For two inequalities, shading the union instead of the overlap.
'The answer must satisfy BOTH. Where do the two shaded regions cross — that overlap is the solution.'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: plotting y = mx + c, solving inequalities and open/closed ends (Lesson 10), and the idea of a region from the Constructions unit. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner draws a solid line for $y > x + 1$, or shades below it. The aim is a question that returns the reasoning.

- "Is > strict or inclusive? So is the boundary dashed or solid?"
- "Test the origin: is 0 > 0 + 1? So which side is the region — the origin's, or the other?"

- “Pick a point in your shaded region and one outside — do they behave as they should?”
- “For two inequalities, where do BOTH hold — which part is shared?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Which test point should they use?

The origin (0, 0) is easiest — substitution is trivial — unless the boundary passes through it, in which case choose another clear point such as (1, 0) or (0, 1).

Dashed or solid — how to remember?

'Or equal to' (≤, ≥) includes the boundary → solid; strict (<, >) excludes it → dashed. Reading the inequality aloud settles it.

How does this connect to the Constructions unit?

A region satisfying several conditions is a compound locus — there the boundaries were circles and bisectors; here they are straight lines. The overlap-of-conditions reasoning is identical.

Why end the unit here?

It completes the arc of what a 'solution' can be: a single value, a pair, a whole line, the crossing of two lines, a range, and finally a region — and it opens the door to linear programming, where shaded regions and their overlaps decide the best option under constraints.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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